

UNIT 1 – Introduction to Website Development & JavaScript

Website Development

Definition:

Website development is the process of creating, designing and maintaining websites for the internet.

It includes layout design, coding, testing and publishing web pages.

Website Development Phases

1. Planning – deciding purpose of website
 2. Designing – creating page layout
 3. Development – writing HTML, CSS, JavaScript code
 4. Testing – checking errors
 5. Deployment – uploading website on server
 6. Maintenance – updating website regularly
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Website Authoring Tools

Definition:

Website authoring tools are software used to create and edit web pages.

Examples:

Notepad

Visual Studio Code

Adobe Dreamweaver

Scripting Language

Definition:

A scripting language is a programming language used to make web pages interactive and dynamic.

Example: JavaScript

Types of Scripting Languages

JavaScript – Client side scripting language

VBScript – Client side scripting language

ASP – Server side scripting language

PHP – Server side scripting language

Difference Between Client Side and Server Side Scripting

Client Side Scripting:

Runs in browser

Fast execution

Reduces server load

Example: JavaScript

Server Side Scripting:

Runs on server

Requires server processing

More secure

Example: PHP, ASP

Introduction to JavaScript

Definition:

JavaScript is a scripting language used to add interactivity and dynamic behavior to web pages.

It works inside web browser.

Advantages of JavaScript

Easy to learn

Fast execution

Creates interactive webpages

Reduces server work

Supported by all browsers

Limitations of JavaScript

Browser dependent
Security limitations
Cannot directly access database
Can be disabled by user

UNIT 2 – Working with JavaScript

Embedding JavaScript into HTML

JavaScript code is written inside script tag.

```
<script>  
document.write("Hello");  
</script>
```

Data Types in JavaScript

Number – numeric values
String – text data
Boolean – true or false
Null – empty value
Undefined – variable without value

Creating Variables

Variables store data values.

```
var a = 10;  
let name = "Rahul";
```

JavaScript Comments

Single line comment:

```
// comment
```

Multi line comment:

```
/* comment */
```

Operators and Expressions

Arithmetic Operators: + - * /

Relational Operators: > < ==

Logical Operators: && ||

Assignment Operator: =

Expression means combination of variables and operators.

Example:

a + b

if-else Statement

Used for decision making.

```
if(age >= 18)
{
    document.write("Eligible");
}
else
{
    document.write("Not Eligible");
}
```

Switch Case Statement

Used when multiple conditions exist.

```
switch(day)
{
    case 1:
        document.write("Monday");
        break;
    default:
        document.write("Invalid");
}
```

Looping Statements

For Loop

```
for(i=1;i<=5;i++)  
{  
  document.write(i);  
}
```

While Loop

```
while(i<=5)  
{  
  i++;  
}
```

UNIT 3 – JavaScript

Functions and Dialog Box

Function

Definition:

Function is a block of code used to perform a specific task.

Creating Function

```
function show()  
{  
  document.write("Hello");  
}
```

Calling Function

```
show();
```

Built-in String Functions

length – returns string length

toUpperCase() – converts to uppercase

toLowerCase() – converts to lowercase

charAt() – returns character position
indexOf() – finds position of character

Example:

```
name.length;
```

Dialog Boxes

Alert Dialog Box

Displays message to user.

```
alert("Welcome");
```

Confirm Dialog Box

Returns true or false.

```
confirm("Are you sure?");
```

Prompt Dialog Box

Takes input from user.

```
prompt("Enter Name");
```

UNIT 4 – JavaScript Events and Objects

JavaScript Events

Definition:

Events are actions performed by user on webpage.

Important Events

onclick – mouse click
onmouseover – mouse over element
onmouseout – mouse leaves element
onkeypress – key pressed
onkeydown – key down
onkeyup – key released
onfocus – input field active
onblur – input field inactive
onload – page loaded
onunload – page closed
onsubmit – form submission

Example:

```
<button onclick="show()">Click</button>
```

String Methods

length()
substring()
replace()
indexOf()
toUpperCase()
toLowerCase()

Math Object

Used for mathematical calculations.

Examples:

```
Math.sqrt(25);  
Math.round(4.5);  
Math.random();  
Math.pow(2,3);
```

Form Object

Definition:

Form object is used to access and control form elements in JavaScript.

Example:

```
document.forms["form1"];
```

Access textbox value:

```
document.form1.name.value;
```